

Unit ID: 679b1a14aaf593b2823f35f3

Unit 4 - Creating Color-Changing LED Lights

PDF Documents

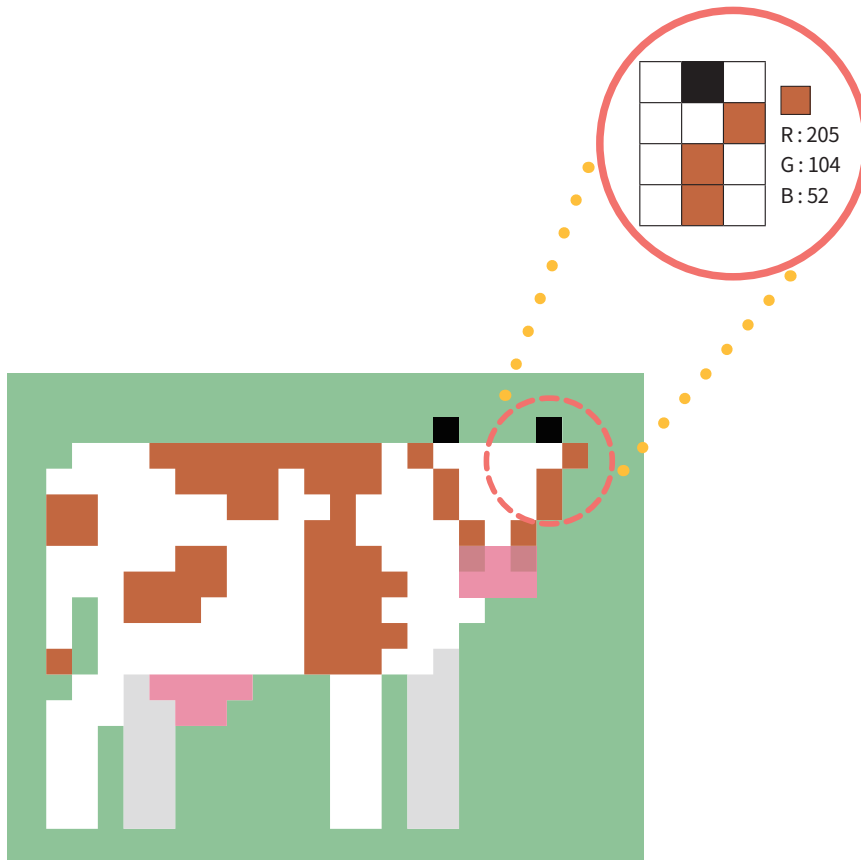
PDF Document 1: 1738218003445-Chapter 5 - Unit 4 - Creating Color-Changing LED Lights.pdf

This PDF document is included in the unit materials.

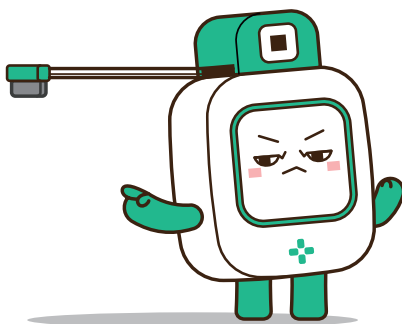
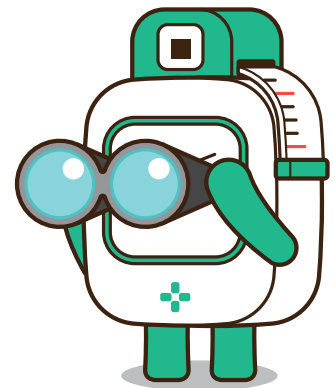
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UNIT 3 CREATING COLOR-CHANGING LED LIGHTS

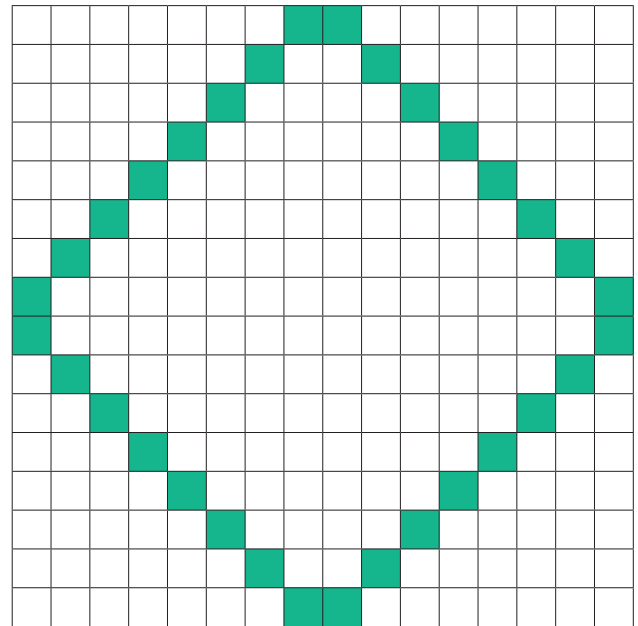
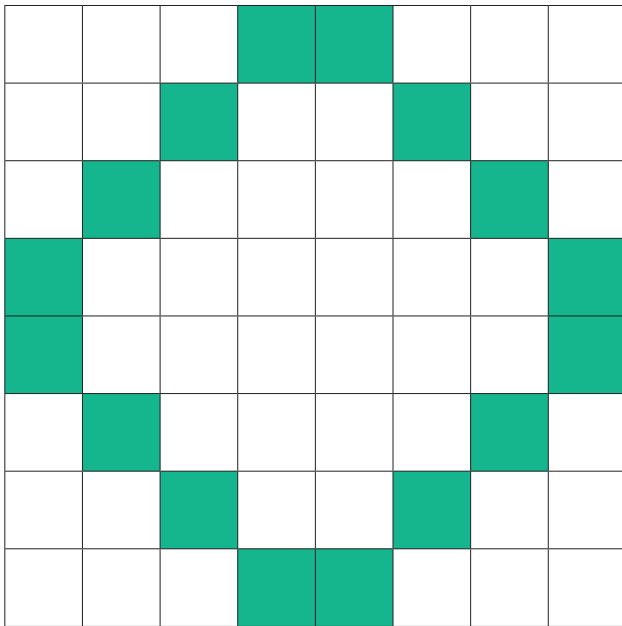


An image is created through a combination of pixels (small dots). If you look closely at a computer, TV, or smartphone screen, you'll see that it comprises these pixels. Each pixel combines different colors and brightness levels to form the images we see.

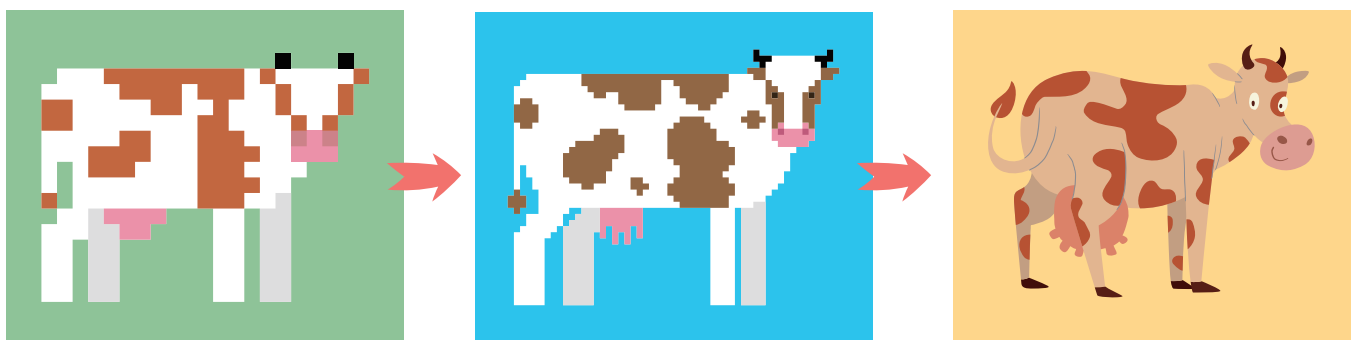
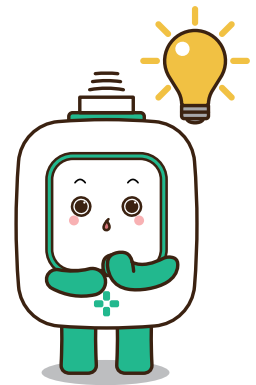




As the number of pixels increases, the image becomes sharper.



An 8x8 pixel grid contains 64 pixels, which we call "64 pixels of resolution." The higher the resolution, the more pixels are included in the image, allowing for sharper and more detailed representation.



This principle displays all the images we see on TV or computer screens.

Multiple Choice Questions

1 In the RGB color model, what color is created by mixing red and blue?

- ☐ (A) Yellow
- ☐ (B) Cyan
- ☐ (C) Purple
- ☐ (D) White

CORRECT ANSWER

(C)

2 Which places would be most suitable for cool-colored lighting?

- ☐ (A) Living room
- ☐ (B) Bedroom
- ☐ (C) Hospital
- ☐ (D) Restaurant

CORRECT ANSWER

(C)

3 What is the basic unit for creating images on a computer screen?

- ☐ (A) Wavelength
- ☐ (B) Waveform
- ☐ (C) Primary colors of light
- ☐ (D) Pixel

CORRECT ANSWER

(D)



Reflection Questions

1 How do you create the color white?

2 Where could the LED light we created be proper?

3 What could improve the sharpness of an image?